System Validation Report

Project: Online Quiz System
Test period: 2 November 2026 to 6 November 2026
Validation result: Accepted for the implemented scope

1. Validation objective

System validation confirms that the implemented system satisfies the user-visible functional, security, and resilience requirements described in the SRS and test plan.

2. Requirements validation

Requirement / feature	Validation method	Result
OQS-F-001 Registration	Valid account, duplicate account, and empty-input tests	Pass
OQS-F-002 Authentication	Valid credentials, invalid credentials, unknown user, and empty-input tests	Pass
OQS-F-003 to OQS-F-005 Quiz catalogue and selection	Valid files, malformed rows, unknown quiz, and empty ID tests	Pass
OQS-F-006 to OQS-F-010 Quiz workflow and scoring	Question parsing, answer validation, complete/incomplete scoring tests	Pass
OQS-F-013 Previous results	Empty submission file and result display test	Pass
OQS-F-014 Leaderboard	Empty submission file and leaderboard display test	Pass
OQS-F-015 Administrator management	Add, edit, delete, duplicate, invalid, and unknown-quiz tests	Pass
OQS-SR-001 Network resilience	Offline server submission test	Pass; returns false without crashing
OQS-SR-002 Password protection	Registration record inspected for absence of plaintext password	Pass
Security logging	Failed login, administrator action, injection, and unwritable-file tests	Pass
OQS-NF-001 Performance	Sprint 2 manual measurement recorded 4 ms response time	Pass against the 3-second target
OQS-NF-002 Usability	Manual console menu and error-message review	Pass

3. Acceptance criteria

- Core user flows execute without crashes.
- Invalid and boundary input is rejected.
- Passwords are not stored in plaintext.
- Security log fields cannot create forged log records.
- Results and leaderboard handling tolerate an empty submissions file.
- Administrator operations preserve quiz-file consistency.

All criteria were satisfied by the available automated and manual evidence.

4. Known limitations

The clean rebuild of all OpenSSL-dependent tests requires a MinGW-compatible OpenSSL installation. Existing compiled security tests passed, but the release build should be repeated in the intended MSYS2 UCRT64 environment.

5. Final decision

System validation: PASS for the tested implementation and documented scope.